Table 24: Prompt used to compare two ideas

```
You are a judge in a competition. You have to decide which idea is
better.

The idea0 is: [idea0]
The idea1 is: [idea1]
The topic is: [topic]
Which idea do you think is better? Please write a short paragraph
to explain your choice.
Here are your evaluation criteria:
1. Novelty: Are the problems or approaches new? Is this a novel
combination of familiar techniques? Is it clear how this work
differs from previous contributions? Is related work adequately
referenced?
2. Significance: Are the idea important? Are other people
(practitioners or researchers) likely to use these ideas or build
on them? Does the idea address a difficult problem in a better way
than previous research? Does it provide a unique theoretical or
pragmatic approach?
3. Feasibility: Can the idea be realized with existing technology
or methods? Are there any technical difficulties or bottlenecks?
Is the idea clear and logical? Is there any obvious error or
unreasonable part in the idea, and can the experiment be designed
normally according to this idea.
4. Clarity: Is the paper clearly written? Is it well-organized?
Does it adequately inform the reader?
5. Effectiveness: How likely the proposed idea is going to work
well (e.g., better than existing baselines).
Note:
Avoid any position biases and ensure that the order in which
the responses were presented does not influence your decision.
DO NOT allow the LENGTH of the responses to influence your
evaluation, choose the one that is straight-to-the-point instead
of unnecessarily verbose. Be as objective as possible. (very
important!!!)
If you think idea0 is better than idea1, you should output 0. If
you think idea1 is better than idea0, you should output 1. If you
think idea0 and idea1 are equally good, you should output 2.

Your output should be strictly in following format:
Your thinking process: ...
Your choice:
Novelty: 0/1/2
Significance: 0/1/2
Feasibility: 0/1/2
Clarity: 0/1/2
Effectiveness: 0/1/2
```

Table 25: Prompt used to compare two experiments

```
You are a judge in a competition. You have to decide which
experiment is better.

The idea of experiment0 is: [idea0]
The experiment0 is: [experiment0]
The idea of experiment1 is: [idea1]
The experiment1 is: [experiment1]
Which experiment do you think is better? Please write a short
paragraph to explain your choice.
Here are your evaluation criteria:
1. Feasibility: Can the experiment be realized with existing
technology or methods? Are there any technical difficulties or
bottlenecks? Is the experimental plan detailed and feasible? Are
the experimental steps clear and logical? Is there any obvious
error or unreasonable part in the experiment. Consider the
rationality of its steps and the possibility that the idea can be
successfully implemented.
2. Quality: Is there a clear rationale for each step of the
experimental design? Are the baseline and evaluation metrics
chosen appropriately? Has the design taken into account the
potential advantages and limitations of the methods used? Can
this experimental design effectively support the claims made in the
idea.
3. Clarity: Is the experimental plan clearly written? Dose it
provide enough information for the expert reader to understand the
experiment? Is it well organized? Does it adequately inform the
reader?
Note: Avoid any position biases and ensure that the order in which
the responses were presented does not influence your decision.
DO NOT allow the LENGTH of the responses to influence your
evaluation, choose the one that is straight-to-the-point instead
of unnecessarily verbose. Be as objective as possible. (very
important!!!)
If you think experiment0 is better than experiment1, you should
output 0. If you think experiment1 is better than experiment0,
you should output 1. If you think experiment0 and experiment1 are
equally good, you should output 2.

Your output should be strictly in following format:
Your thinking process: ...
Your choice:
Feasibility: 0/1/2
Quality: 0/1/2
Clarity: 0/1/2
```

Table 26: Evaluation results of idea generation for both model-based evaluation and human-based evaluation.

		Novelty	Significance	Clarity	Feasibility	Effectiveness	Average	Rank
Human	Real Paper	<u>1075</u>	<u>1071</u>	**1118**	**1127**	**1109**	**1100**	**1**
	CoI Agent (ours)	**1100**	**1103**	<u>1081</u>	<u>1065</u>	<u>1078</u>	<u>1085</u>	<u>2</u>
	RAG	1021	1038	1022	1030	1035	1029	3
	GPT-Researcher	988	993	993	990	999	992	4
	ResearchAgent	982	975	1001	975	970	980	5
	AI-Scientist	835	820	784	813	809	812	6
GPT-4o	Real Paper	<u>1063</u>	<u>1089</u>	**1137**	**1165**	<u>1123</u>	**1115**	**1**
	CoI Agent (ours)	**1144**	**1138**	<u>1080</u>	1021	**1152**	<u>1107</u>	<u>2</u>
	GPT-Researcher	995	1007	995	1010	989	999	3
	ResearchAgent	1005	1016	1005	946	1004	995	4
	RAG	914	918	978	<u>1023</u>	918	950	5
	AI-Scientist	878	831	806	836	814	833	6
Gemini1.5-Pro-Exp0827	Real Paper	<u>1102</u>	<u>1101</u>	**1125**	**1120**	<u>1102</u>	**1110**	**1**
	CoI Agent (ours)	**1124**	**1119**	<u>1098</u>	<u>1082</u>	**1113**	<u>1107</u>	<u>2</u>
	GPT-Researcher	1002	997	1005	1014	998	1003	3
	ResearchAgent	986	986	984	975	986	983	4
	RAG	914	929	948	958	932	936	5
	AI-Scientist	873	868	840	851	869	860	6
Claude-3.5-Sonnet	Real Paper	<u>1099</u>	<u>1123</u>	**1174**	**1149**	**1179**	**1145**	**1**
	CoI Agent (Ours)	**1165**	**1154**	1033	953	<u>1162</u>	<u>1094</u>	<u>2</u>
	GPT-Researcher	986	977	1022	1039	977	1000	3
	ResearchAgent	1008	1023	<u>1034</u>	926	997	998	4
	RAG	886	907	977	<u>1038</u>	884	938	5
	AI-Scientist	855	815	760	895	800	825	6